

BrokerVerse OOTB

Insurance broking platform by iorta TechNXT

Solution Architecture · Document 10 of 11

Data Archival, Housekeeping and Restoration

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Version	Date	Author	Change
1.0	29 September 2026	iorta TechNXT	Initial issue
1.1	29 September 2026	iorta TechNXT	Updated to the current system: placement journey, co-insurance, posting rules, period end and BIR tax, bank reconciliation, document numbering, commission rate matrix, single PDF engine, uploads and go-live imports, housekeeping, business time zone, broker roles, deployment on the BrokerVerse URL
1.2	04 October 2026	iorta TechNXT	New modules (AML/CFT, BIR, payables and fixed assets, distribution, fleet and marine, My Work, go-live workbench), integration framework, encryption and masking of personal data, branding and e-signatures, release pipeline; database at migration 0322 (reference data), new ER diagrams

Review and approval

Role	Name	Signature	Date
Solution architect			
DevOps / infrastructure lead			
Information security			
Business owner			

Conventions

- Statements of fact are derived from the code, configuration and database of the baseline above (file paths are given relative to the repository). Figures measured on the local test system are labelled **test data** or **indicative**.
- Targets, sizes, retention periods and thresholds that are not facts of the code are labelled **Recommended / to be confirmed by the business and DevOps**.

- Items marked **Gap** are defects or missing capabilities found while preparing this document.

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1 Introduction

1.1 Purpose and scope

This document describes how BrokerVerse OOTB data grows, the housekeeping the system performs, the recommended retention periods, the approach to archiving old records out of the online database, how archived data is restored on request, and how the sample-data purge script is used before go-live.

Recommended / to be confirmed by the business and DevOps: Retention periods in this document are proposals. The compliance officer must confirm them against the rules that apply to the broker, in particular the record-keeping rules of the Bureau of Internal Revenue (BIR) and the Insurance Commission (IC), and the Data Privacy Act of 2012 (personal data kept no longer than needed for the declared purposes). This document gives the regulatory context in general terms only; it is not legal advice.

2 What grows

Data	Growth driver	Handling in the baseline
Business and financial records (about 190 tables: leads to journals, placement, period end, bank reconciliation, payables, fixed assets, BIR filings and invoices, AML records, compliance registers)	Every quotation, policy, receipt, voucher, claim, statement line	Kept indefinitely; financial documents are cancelled or reversed, not deleted; draft quotations can be deleted; leads, quotations and bank statements have <code>deleted_at</code>
<code>audit_log</code>	Every change, sign-in related event, job run started by a user, report generation	Kept; purged only if <code>housekeeping.audit_log_days</code> is set, never below seven years
<code>login_history</code>	Every password, two-factor and refresh attempt	Purged by housekeeping after 365 days
<code>job_runs</code>	About 2,550 rows a day with the jobs enabled by default: <code>renewal-queue</code> every minute (1,440), <code>integration-outbox</code> every 2 minutes (720), <code>email-outbox</code> every 5 minutes (288), <code>my-work-reminders</code> every 15 minutes (96) and the daily jobs	Purged by housekeeping after 90 days
<code>notifications</code>	Reminders, approvals, renewal and collection notices	Read notifications purged after 180 days; users can delete them in the bell menu
<code>email_outbox</code>	Every e-mail (full HTML body kept)	Sent messages purged after 180 days, failed ones after 730 days
<code>refresh_tokens</code>	Every sign-in and every refresh (rotation)	Expired or revoked rows purged 30 days after expiry or revocation
<code>password_resets</code>	Forgot-password requests	Used or expired codes purged after 7 days
<code>password_history</code>	Password changes	Trimmed to the last <code>security.password_history_count</code> (5) per user by <code>lib/password.js</code>
<code>job_queue</code>	Renewal batch notices	Completed items purged after 30 days
<code>generated_reports</code> and report files	Scheduled daily reports and user-generated reports	Rows and files older than <code>reports.retention_days</code> (90) deleted by the <code>daily-reports</code> job
Uploaded files (<code>UPLOAD_DIR</code>)	Documents, photos, statements, bordereaux, exports	No lifecycle; deleted only through the delete endpoint (uploader, module writer or administrator)
<code>integration_outbox</code> , <code>integration_attempts</code> , <code>integration_inbox</code>	Every SMS, Viber, CTPL, insurer and bank file message and each attempt	Not purged (Gap)

Data	Growth driver	Handling in the baseline
eis_submissions, aml_provider_requests	Every sales invoice sent to the BIR EIS; screening provider requests	Not purged (Gap); AML records kept for <code>aml.record_retention_years</code>
data_load_rows	Every row of every uploaded go-live workbook	Not purged (Gap); needed only until the load is reconciled
Status housekeeping	Policies past expiry, quotations past validity, renewals past grace	Jobs <code>policy-expiry</code> , <code>quote-expiry</code> , <code>renewal-pipeline</code> change the status (no deletion)

Note: In production, files without a database record (orphans) can arise from failed uploads (a key is reserved before the file is written) and from restores done at different times for the database and the files. A periodic reconciliation report is recommended (chapter 4).

3 Retention policy

3.1 Regulatory context

- **BIR:** books of accounts and their supporting documents (official receipts, invoices, vouchers, journals, withholding tax certificates such as BIR Form 2307) must be kept for the period set by the revenue regulations, currently up to ten years from the date of the last entry (Revenue Regulations No. 17-2013 as amended), and be available for examination.
- **Insurance Commission:** insurance brokers must keep complete records of the business transacted (policies placed, premiums received and remitted, commissions) available for examination, for the period required by the Insurance Code and IC circulars.
- **Data Privacy Act of 2012 (Republic Act 10173):** personal data of leads, clients and claimants may be kept only as long as needed for the purpose or as required by law, then disposed of securely; access and retention must be documented.

3.2 Recommended retention periods

Class / data (Recommended / to be confirmed by compliance)	Tables	Online (PostgreSQL)	Total, incl. archive
Financial records	receivables and credits, receipts and applications, journal vouchers and lines, entry matches, disbursements, invoice lists, cheques, petty cash, commissions and adjustments, remittances, allocations and refunds, debit notes and collections, direct-bill items, claim settlement cash, bank statements, matches and reconciliations, close runs, year-end runs, opening balances, BIR 2307 certificates, reinsurance recoveries and reconciliations, incentive calculations and results, policy payments	Current year + 2 closed years	10 years after the end of the financial year
Policy and claim records	clients, quotations (converted), broker slips, offers, placements, co-insurance participants, policies, endorsements, claims with history, renewals, cessions, treaties, bordereaux, documents	Life of the policy or claim + 2 years	10 years after policy expiry or claim closure
Leads and quotations not converted	leads, quotes (not converted), broker slips without placement, renewal quotes, win-back campaigns	2 years	3 years, then anonymise or delete (privacy)
Audit trail	audit_log, claim_field_changes, claim_history, collection_actions, renewal_activities, renewal_notices, remittance_approvals, period_status_history, bank_reconciliation_history	2 years	10 years (supports the financial records)
Security logs	login_history, password_resets	1 year	2 years

Class / data (Recommended / to be confirmed by compliance)	Tables	Online (PostgreSQL)	Total, incl. archive
Operational logs	job_runs	90 days	1 year (aggregated)
Temporary data	refresh_tokens (expired or revoked), email_outbox (sent), notifications (read), job_queue (done), generated_reports	30 to 180 days	Same (no archive)
Reference and configuration	settings, masters, posting rules, number series, tax codes, product templates, roles, report catalogue	While in use	Versions kept through the audit log
Uploaded files	UPLOAD_DIR	As the owning record	As the owning record

4 Housekeeping

4.1 Housekeeping job

The scheduled job `housekeeping` (daily at 02:45 Manila time, `jobs/housekeeping.js`) deletes operational rows past their retention period. Each period is a setting in System Settings > Housekeeping (days; 0 keeps the rows forever). Deletes run in batches of 5,000 rows so a large backlog never holds long locks, and the job output lists the rows deleted per table and the tables kept.

Setting	Rule	Default days	Supporting index
<code>housekeeping.job_runs_days</code>	<code>job_runs</code> started before the cut-off	90	<code>job_runs (job_id, started_at)</code>
<code>housekeeping.email_outbox_sent_days</code>	<code>email_outbox</code> with status sent	180	<code>email_outbox (status, created_at)</code>
<code>housekeeping.email_outbox_failed_days</code>	<code>email_outbox</code> with status failed	730	<code>email_outbox (status, created_at)</code>
<code>housekeeping.login_history_days</code>	<code>login_history</code>	365	<code>login_history (at DESC)</code>
<code>housekeeping.refresh_tokens_days</code>	<code>refresh_tokens</code> expired or revoked before the cut-off	30	primary key and user index
<code>housekeeping.password_resets_days</code>	<code>password_resets</code> used or expired before the cut-off	7	primary key
<code>housekeeping.notifications_read_days</code>	<code>read notifications</code>	180	<code>notifications (user_id, is_read, created_at)</code>
<code>housekeeping.job_queue_done_days</code>	<code>job_queue</code> items with status completed	30	primary key
<code>housekeeping.audit_log_days</code>	<code>audit_log</code> older than the cut-off, never less than 2,557 days (seven years)	0 (kept)	<code>audit_log (at)</code>

4.2 Other housekeeping

Job or rule	Schedule (Manila time)	Effect
daily-reports	05:00	Generates the daily reports and deletes generated reports and their files older than <code>reports.retention_days</code>
policy-expiry	00:15	Status <code>expired</code> for policies past expiry
quote-expiry	00:30	Status <code>expired</code> for quotations older than <code>limits.quote_validity_days</code>
renewal-pipeline	05:30	Lapses renewals past <code>renewals.grace_period_days</code>
cover-note-expiry	06:20	Status <code>expired</code> for cover notes past their validity
Data subject request, anonymisation	on request	Personal data of a client or prospect anonymised once <code>privacy.retention_years</code> has passed (Master > Data Privacy)
Transaction reset	before go-live	<code>npm run reset:transactions</code> removes the smoke-test records; refused once <code>golive.locked</code> is on
Password change	on change	Keeps only the last N password hashes

4.3 Recommended additions

Gap: Housekeeping has no retention rule yet for the integration outbox, attempts and inbox, the EIS outbox, the screening provider requests and the go-live workbook rows. **Recommended:** 180 days for sent messages and

attempts, 730 days for failed ones, 90 days for workbook rows after the load is reconciled.

Recommended / to be confirmed by the business and DevOps: Archive `login_history` and, once the compliance period is agreed, `audit_log` before housekeeping deletes them (chapter 5). Add a weekly report of files in `UPLOAD_DIR` without a `documents` or `generated_reports` row, and of rows whose file is missing; do not delete automatically.

5 Archival

5.1 Approach

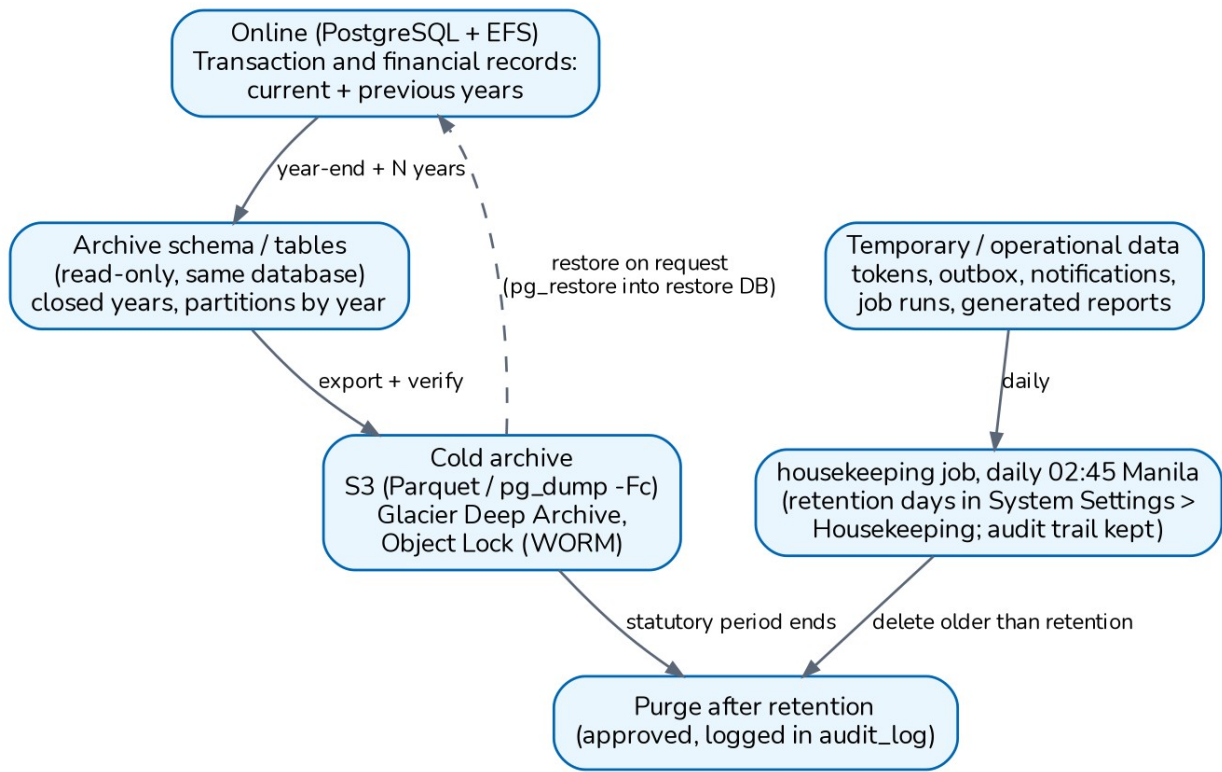


Figure 1. Data lifecycle: online, archive, purge (recommended)

Stage	Approach (Recommended / to be confirmed by the business and DevOps)
Online	Operational data stays in the main tables for the online period above; reports and screens work unchanged.
Archive in the database	For the high-volume tables (<code>audit_log</code> , <code>login_history</code> , later <code>journal_lines</code> and <code>bank_statement_lines</code>), convert to tables partitioned by year (declarative partitioning on <code>at</code> or <code>jv_date</code>) when the volume justifies it (for example above 10 million rows); old partitions can then be detached, exported and dropped in seconds without long deletes. Until then, move rows older than the online period into <code>archive.<table></code> tables with the same structure in a separate schema, in batches.
Cold archive	Once a year, after the financial year is closed (year-end run) and audited: export the closed year (per table, filtered by date or by policy or financial year) with <code>COPY ... TO</code> as CSV or Parquet, or as a <code>pg_dump -Fc</code> of the archive schema, to an S3 bucket with Object Lock (compliance mode) and a lifecycle to S3 Glacier Deep Archive. Record a manifest (tables, row counts, checksums, date range, migration level from <code>schema_migrations</code>). Copy the files of the archived records from the file store to the same bucket under their storage keys.
Purge	Only after the export is verified (row counts and checksums) and approved by the data owner: delete the archived rows from the online database and record the purge in <code>audit_log</code> . Financial records are purged only after the statutory period.
Referential integrity	Archive whole business units together (a policy with its quotation, placement, bills, receipts, journals, commissions, claims and documents) or keep the referenced parent rows online; foreign keys (<code>NO ACTION</code>) prevent deleting a parent that is still referenced. Opening balances of later years keep the ledger totals correct after old journals are archived.

The document-number counters (sequences) stay online; old periods are small.

5.2 Restoration from archive

1. The data owner requests the records (for example a policy file of a given year for an audit or a claim dispute); the request is approved and logged.
2. DevOps restores the archive object from Glacier (standard retrieval up to 12 hours, bulk up to 48 hours for Deep Archive).
3. The data is loaded into a separate restore database, never into production: `pg_restore` of the archive dump, or `CREATE TABLE` and `COPY ... FROM` of the CSV or Parquet files, at the migration level recorded in the manifest.
4. The requester gets read access through a read-only database login or a report extract; the files come from the restored S3 objects.
5. The restore database is deleted once the purpose is fulfilled; the access is recorded.

If records must come back into production (for example a reopened claim), load them with the same ids after a review of the dependent rows. Ids are random text or serial values and do not collide with new records, and the document-number counters are not affected.

6 Sample-data purge before go-live

`backend/scripts/purge-sample-data.js` (`npm run purge:sample`) removes demo data from a database that was started with `SEED_SAMPLE_DATA` on (for example a UAT database promoted to production). It is a one-off go-live tool, not a housekeeping job.

Aspect	Behaviour
Safety	Refuses to run without <code>CONFIRM_PURGE=yes</code> ; dry run by default (row counts per table); <code>--execute</code> deletes, in one transaction (any error rolls everything back). Stop the API instances first.
Removes	All rows of the transaction tables (leads to journals, broker slips, offers, placements and co-insurance participants, claims, renewals, fiscal years, opening balances, close runs, recurring journals, bank statements and reconciliations, BIR 2307 certificates, notifications, documents and generated-report records, job history, e-mail outbox, document-number counters), the sample master rows by natural key (fictional insurers, demo branches, signatories, referrers, reinsurers, incentive programmes, demo master records such as bank accounts and commission rates) and the six sample users. With <code>--keep-users=a,b</code> , every user except <code>BrokerVerse</code> and those listed.
Keeps	Reference data (settings, roles, permissions, scheduled jobs, masters, chart of accounts, posting rules, number series, tax codes, product templates, report catalogue) and the audit trail (the purge is recorded in it); <code>--purge-audit</code> also empties <code>audit_log</code> and <code>login_history</code> .
Files	Uploaded files of removed records stay in <code>UPLOAD_DIR</code> ; clear the directory if nothing real was uploaded.
Afterwards	Set <code>SEED_SAMPLE_DATA=false</code> (or remove it) before starting the API, otherwise the sample data is seeded again. Then load the go-live data: masters, users, opening balances and open items (<code>docs/onboarding/GO_LIVE_DATA_SETUP.md</code> and the templates in <code>docs/package/05_Delivery/Upload_Templates</code>).

Procedure (`deploy/README.md`, section 7): take a snapshot; stop the API; `CONFIRM_PURGE=yes npm run purge:sample` and review the counts; `CONFIRM_PURGE=yes npm run purge:sample -- --execute [--keep-users=...]`; set `SEED_SAMPLE_DATA=false`; start the API; check that Leads, Clients and Policies are empty.

7 Related documents

This document is one of eleven that together form the BrokerVerse OOTB Solution Architecture. They are maintained and issued together (source and build: [docs/architecture/README.md](#)).

No.	Document	File
01	Solution Component Diagram and Application Architecture	01_BrokerVerse_Solution_Component_Diagram_Application_Architecture.pdf
02	Database Design	02_BrokerVerse_Database_Design.pdf
03	Database Inventory	03_BrokerVerse_Database_Inventory.pdf
04	Technology Stack	04_BrokerVerse_Technology_Stack.pdf
05	Shared Service Components	05_BrokerVerse_Shared_Service_Components.pdf
06	Capacity and Performance	06_BrokerVerse_Capacity_and_Performance.pdf
07	High Availability and Resiliency	07_BrokerVerse_High_Availability_and_Resiliency.pdf
08	RTO and RPO	08_BrokerVerse_RTO_and_RPO.pdf
09	Backup and Recovery	09_BrokerVerse_Backup_and_Recovery.pdf
10	Data Archival, Housekeeping and Restoration (this document)	10_BrokerVerse_Data_Archival_Housekeeping_and_Restoration.pdf
11	Monitoring	11_BrokerVerse_Monitoring.pdf

8 Glossary

Terms and abbreviations used in this document.

Term	Meaning
API	Application programming interface; here the BrokerVerse REST API under /api served by the Node.js backend.
Audit log / audit trail	Table audit_log: who changed which record, when, from which IP, with the before and after values.
BIR	Bureau of Internal Revenue (Philippines): tax authority whose rules drive record keeping and withholding tax certificates.
BIR Form 2307	Certificate of creditable tax withheld at source, issued by the payor to the payee each quarter.
Co-insurance	Placement of one risk with several insurers, one of them the lead, each taking a share; premium, commission and claims are split by share.
CTPL	Compulsory Third Party Liability motor insurance (Philippines).
Housekeeping	Daily job that deletes operational rows (job runs, sent e-mails, sign-in history, expired tokens, read notifications) after the retention days in System Settings.
IC	Insurance Commission (Philippines): regulator of insurers and brokers.
Migration	Numbered SQL file in backend/src/db/migrations applied once, in order, on API start.
Receivable / bill	Premium bill (INV-) raised when a policy, endorsement or renewal is issued.
Remittance	Payment of collected premium, net of commission, from the broker to the insurer.
RPO	Recovery point objective: the maximum acceptable data loss, measured in time.
RTO	Recovery time objective: the maximum acceptable time to restore service.
S3	Amazon Simple Storage Service: object storage hosting the SPA build (and proposed archive store).
Seed	Idempotent data load run on every API start: reference data always, sample data only when SEED_SAMPLE_DATA is on.
UAT	User acceptance testing environment.